Fiscal Year:	FY 2026	**Task Last Updated:**	FY 10/24/2025
PI Name:	Zhen, Shuyang Ph.D.		
Project Title:	Investigating the Impact of Foliar and Root-Zone Exposure to Lunar Regolith Simulant on Lettuce Growth and Stress Physiology in a Hydroponic System		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	shuyang.zhen@tamu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	207-561-0894
Organization Name:	Texas A&M AgriLife Research		
PI Address 1:	495 Horticulture Road		
PI Address 2:	2133 TAMU, Room 202		
PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77845	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2022 Space Biology NNH22ZDA001N-SBR: E.9 Space Biology Research Studies
Start Date:	01/01/2024	**End Date:**	05/31/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Mickens, Matthew	**Contact Phone:**	
Contact Email:	Matthew.A.Mickens@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 05/31/2026 per the PI (Ed., 10/29/25). NOTE: End date changed to 09/30/2025 per the PI (Ed., 5/1/25).		
Key Personnel Changes/Previous PI:	There are no changes to key personnel.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC24K0812		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Fresh food production, using in situ resources such as lunar regolith, is crucial for sustaining long-duration human explorations on the Moon. However, lunar regolith has vastly different physical, chemical, mineralogical, and hydraulic properties compared to terrestrial soils. The suitability of lunar regolith as a substrate for crop growth remains largely untested and poorly understood. On the other hand, hydroponic crop cultivation has been extensively researched and widely utilized in controlled environment agriculture on Earth. It readily allows for root-zone control and optimization, thus making it a potentially more predictable and reliable growing method for future Moon colonies. This one-year plant-focused Early Career Research Investigation aims to evaluate the suitability and potential toxicity of two lunar regolith simulants, OPRH4W30 (representing lunar highland regolith) and JSC-1A (representing lunar mare regolith), as substrates for plant growth. Specifically, we will quantify the responses of food crops to these simulants and compare them to hydroponics. Our objectives are to: (1) compare seed germination and plant growth of lettuce (Lactuca sativa) in lunar regolith simulants, with or without supplementary nutrients, to hydroponically grown lettuce during the seedling stage; (2) quantify the impact of root-zone exposure to lunar regolith simulants on the growth and stress physiology of hydroponically grown lettuce; and (3) quantify the impact of foliar exposure to lunar regolith simulants on the growth and stress physiology of hydroponically grown lettuce. Two lettuce cultivars, ‘Rex’ (a green leaf butterhead lettuce) and ‘Outredgeous’ (a red romaine lettuce) will be used for this investigation. Lettuce is a suitable pick-and-eat leafy vegetable for a crew dietary supplement. For objective 1, we will germinate and grow plants in five different substrates/fertilization treatments: two types of regolith simulants (OPRH4W30 and JSC-1A) each with or without supplementary nutrients, and a control treatment where plants will be grown hydroponically in an inert rockwool substrate with a complete nutrient solution. For objectives 2 and 3, we will evaluate the impact of root-zone and foliar exposures to regolith simulants on hydroponically grown lettuce by implementing several of the following treatments, including a control treatment with no foliar or root-zone exposure to lunar regolith simulants; root-zone exposure to regolith simulants OPRH4W30 and JSC-1A at a low and high concentration; foliar exposure to dilute OPRH4W30 or JSC-1A solution at 2-d or 4-d intervals; and foliar spray with deionized water at 2-d intervals (as another control for the foliar exposure treatments). All plants will be grown in a climate-controlled walk-in growth room under light emitting diode lights. Seed germination, plant growth yield, beneficial phytochemical contents, and physiological stress indicators -- such as the maximum quantum efficiency of photosystem II (PSII) (Fv/Fm) -- will be assessed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will establish baseline seed germination, plant growth, morphological and phytochemical quality attributes, and stress responses of lettuce grown directly in lunar regolith simulants or in contact with regolith simulants. The results of this project will provide important insight on the suitability and potential toxicity of lunar regolith (versus hydroponics) as a substrate for in situ food production on the Moon.
Task Progress:	We have completed the first of the two proposed studies and the plant growth experiments for study 2. The analysis of plant tissue elemental composition to detect potential nutrient disorders (study 2) is still in progress. These analyses help provide further insights into the effects of regolith exposure on hydroponic lettuce growth. Key results: Study 1. We found lettuce seedlings grown in lunar regolith with deionized water exhibited severely stunted growth. In contrast, nutrient supplementation significantly improved seedling growth, but still resulted in substantially smaller plants compared to control treatments with commercial soilless substrates and nutrient supplementation. Study 2. Growth parameters, including total leaf area and shoot fresh and dry biomass, did not differ significantly among treatments with different concentrations or frequencies of root zone or foliar exposure to regolith simulants. These results suggest that neither root zone or foliar exposure to lunar regolith simulants at the tested concentrations induced adverse effects on plant growth. This finding is encouraging, indicating that hydroponic plant production may be minimally affected by relatively low levels of lunar regolith exposure, which is likely unavoidable in a future lunar habitat.
Bibliography Type:	Description: (Last Updated:)